Optimal Optimistic Reliable Broadcast

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1 Optimistic Reliable Broadcast

Claim 1. *If an honest party optimistically commits m, then no honest party will send* vote *or* ack *for any value $m' \neq m$.*

Proof. For an honest party P_j to send vote on some another value m', it must receive $\langle \mathsf{echo}, m' \rangle$ from at least $\lceil \frac{n}{2} \rceil$ non-broadcaster parties which, since $\lceil \frac{n}{2} \rceil > f$, is impossible when the broadcaster is honest. Thus, we consider the case where the broadcaster is Byzantine. In this case, there are at most $(f-1)$ Byzantine non-broadcaster parties.

Assume that an honest party P_i optimistically commits to the value m. This indicates that P_i must have received $\langle \mathsf{echo}, m \rangle$ from at least $\lceil \frac{n+2f-2}{2} \rceil$ distinct non-broadcaster parties. Therefore, at least $\lceil \frac{n+2f-2}{2} \rceil - (f-1) = \lceil \frac{n}{2} \rceil$ honest parties must have sent $\langle \mathsf{echo}, m \rangle$. Now, for the sake of contradiction, suppose an honest party P_j sends a vote for another value $m' \neq m$. This would mean that P_j must have received $\langle \mathsf{echo}, m' \rangle$ from $\lceil \frac{n}{2} \rceil$ distinct non-broadcaster parties, of which at least $\lceil \frac{n}{2} \rceil - (f-1) = \lceil \frac{n-2f+2}{2} \rceil$ must have come from honest parties. Observe that $\lceil \frac{n}{2} \rceil + \lceil \frac{n-2f+2}{2} \rceil - (n-f) \geq 1$, implying that at least one honest party must have sent echo for both m and m'. This is a contradiction. Hence, no honest party can send a vote for a conflicting value.

Similarly, by a straightforward quorum intersection argument, it is impossible for $\lceil \frac{n+f-1}{2} \rceil$ echo messages for any value $m' \neq m$ to exist. Consequently, no honest party will send an ack message for m' upon receiving $\lceil \frac{n+f-1}{2} \rceil$ echo messages for m'.

Furthermore, since no honest party sends a vote for m', no honest party can receive $\lceil \frac{n+f-1}{2} \rceil$ vote messages for m'. Therefore, no honest party will send an ack for m'. □

Claim 2. *If an honest party optimistically commit value m, all other parties will commit value m.*

Proof. Assume that an honest party P_i optimistically commits to the value m. This implies that P_i must have received $\langle \mathsf{echo}, m \rangle$ from at least $\lceil \frac{n+2f-2}{2} \rceil$ non-broadcaster parties. Therefore, at least $\lceil \frac{n}{2} \rceil$ honest non-broadcaster parties must have sent $\langle \mathsf{echo}, m \rangle$ to all parties, and thus every honest party eventually receives at least $\lceil \frac{n}{2} \rceil$ $\langle \mathsf{echo}, m \rangle$ messages.

By Claim 1, no honest party sends a vote for any $m' \neq m$. Consequently, all honest parties send $\langle \mathsf{vote}, m \rangle$ to all parties and eventually receive at least $\lceil \frac{n+f-1}{2} \rceil$ distinct $\langle \mathsf{vote}, m \rangle$ messages. Again, by Claim 1, no honest party sends an ack for $m' \neq m$. Hence, all honest parties send $\langle \mathsf{ack}, m \rangle$ and eventually receive at least $\lceil \frac{n+f-1}{2} \rceil$ $\langle \mathsf{ack}, m \rangle$ messages.

Finally, for an honest party to send ready for some value $m' \neq m$, at least one honest party must receive at least $\lceil \frac{n+f-1}{2} \rceil$ $\langle \mathsf{ack}, m' \rangle$ messages. However, since no honest party sends an ack for m', no honest party sends ready for m'. Moreover, since all honest parties eventually receive at least $\lceil \frac{n+f-1}{2} \rceil$ $\langle \mathsf{ack}, m \rangle$ messages, they will send ready for m and eventually receive at least $2f+1$ $\langle \mathsf{ready}, m \rangle$ messages, ensuring that they commit to the value m. □

Claim 3. *If an honest party sends* ready *for m, then no honest party will send* ready *for any value $m' \neq m$.*

1. **Propose.** The broadcaster P_k sends $\langle \mathsf{propose}, m\rangle$ to all parties.
2. **Echo.** Upon receiving the first $\langle \mathsf{propose}, m\rangle$ message from the broadcaster, party P_i sends $\langle \mathsf{echo}, m\rangle$ to all parties.
3. **Vote.** Upon receiving $\langle \mathsf{echo}, m\rangle$ from $\lceil \frac{n}{2} \rceil$ non-broadcaster parties, send $\langle \mathsf{vote}, m\rangle$ to all parties if not already sent.
4. **Ack.** Party P_i sends a $\langle \mathsf{ack}, m\rangle$ message to all parties (if not already sent) under either of the following conditions:
   - Upon receiving $\langle \mathsf{vote}, m\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties,
   - Upon receiving $\langle \mathsf{echo}, m\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties
5. **Ready.** Party P_i sends a $\langle \mathsf{ready}, m\rangle$ message to all parties (if not already sent) under either of the following conditions:
   - Upon receiving $\langle \mathsf{ack}, m\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties,
   - Upon receiving $f+1$ $\langle \mathsf{ready}, m\rangle$
6. **Opt Commit.** Upon receiving $\langle \mathsf{echo}, m\rangle$ from $\lceil \frac{n+2f-2}{2} \rceil$ non-broadcaster parties, party P_i commits m.
7. **{4,5}-step Commit.** Upon receiving $2f+1$ $\langle \mathsf{ready}, m\rangle$ messages, party P_i commits m and terminates.

Figure 1: (2,4,5) Optimistic reliable broadcast under $n \geq 3f+1$

Proof. For an honest party to send ready for some value m', at least one honest party must have received $\langle \mathsf{ack}, m'\rangle$ from at least $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties. Similarly, for an honest party to send $\langle \mathsf{ack}, m'\rangle$, it must have either (i) received $\langle \mathsf{vote}, m'\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties, or (ii) received $\langle \mathsf{echo}, m'\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties.

In the latter case, $\lceil \frac{n+f-1}{2} \rceil > f$, which implies that at least one honest party must have sent $\langle \mathsf{echo}, m'\rangle$ which is impossible when the broadcaster is honest. In the former case, at least one honest party must have sent $\langle \mathsf{vote}, m'\rangle$, which occurs only after receiving $\lceil \frac{n}{2} \rceil > f$ $\langle \mathsf{echo}, m'\rangle$ messages which is also impossible when the broadcaster is honest. Therefore, it suffices to consider the case where the broadcaster is Byzantine, in which case there are at most $f-1$ Byzantine non-broadcaster parties.

For an honest party P_i to send ready for m, at least one honest party must have received $\langle \mathsf{ack}, m\rangle$ from at least $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties. Suppose, for the sake of contradiction, that an honest party P_j sends ready for some value $m' \neq m$. Then at least one honest party must have received $\langle \mathsf{ack}, m'\rangle$ from $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties.

Therefore, there must exist two sets of at least $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties sending ack for m and m' respectively. Observe that $\lceil \frac{n+f-1}{2} \rceil + \lceil \frac{n+f-1}{2} \rceil - (n-1) \geq (f-1)+1$. Therefore, at least one honest party must have sent ack for both m and m'. A contradiction. □

Theorem 1. *The protocol in Figure 1 solves Byzantine reliable broadcast under asynchrony with optimal resilience $f < n/3$ and optimistic good-case latency of 2 steps, good-case latency of 4 steps and has bad-case latency of 5 steps.*

Proof. **Validity and good-case latency.** When the broadcaster is honest, it proposes the same value m to all parties. Consequently, all honest parties send echo for m, ensuring that every honest party receives at least $\lceil \frac{n+f-1}{2} \rceil$ echo messages for m. This triggers them to send ack for m. Subsequently, all honest parties receive $\langle \mathsf{ack}, m\rangle$ from at least $\lceil \frac{n+f-1}{2} \rceil$ non-broadcaster parties and send ready for m. Once every honest party receives at least $2f+1$ ready messages for m, they commit to m. This protocol therefore achieves a good-case latency of 4 steps.

Furthermore, if at least $\lceil \frac{f-1}{2} \rceil$ other parties send echo for m to at least one honest party, that party can commit m within 2 steps, resulting in an optimistic good-case latency of 2 steps.

Agreement and bad-case latency. We first argue that no two honest parties commit to different values. If two honest parties optimistically commit to different values, then there must be at least $\lceil \frac{n}{2} \rceil + \lceil \frac{n}{2} \rceil - (n-f) \geq f$ honest parties who sent echo for both of these values. This is impossible because honest parties only send

echo for a single value. Similarly, no two honest parties can commit different value using 4 or 5 step commit rule.

Now we show that if an honest party P_i optimistically commits m and another honest party P_j commits m' using 4 step commit rule, then it must be that $m = m'$. By Claim 1, no honest party sends vote or ack for value m'. Thus, no honest party sends ready for m' and no honest party commits value m' using 4 step commit rule. Thus, $m = m'$.

Next, we argue that if an honest party commits value m, all honest parties will commit m. If the broadcaster is honest, by validity, all honest parties will commit the same value. Now consider the case when the broadcaster is Byzantine. When an honest party P_i optimistically commits to m. By Claim 2, all honest parties will commit m.

Now, consider the case where an honest party commits to m upon receiving $2f + 1$ ready messages for m. This implies that at least $f + 1$ honest parties must have sent ready for m, and these messages will eventually be received by all honest parties. Furthermore, by Claim 3, no honest party sends ready for any value $m' \neq m$. Consequently, any honest parties that have not yet sent a ready message will send ready for m. As a result, all honest parties will eventually receive $2f + 1$ ready messages for m and commit m. This processs achieves a bad-case latency of 5 steps.

Integrity. Integrity follows immediately as each honest party commits and outputs a single value. □